

1. Mean Aggregate Spike Rate (R)

First, we find how many total spikes the hardware is generating every second.

Formula = $R = N \times f$

$R = 1024 \text{ neurons} \times 50 \text{ Hz}$

$R = 51,200 \text{ spikes/second}$

2. Mean AER Bandwidth (B)

$B = R \times 20$

$B = 51200 \times 20 = 1024000 \text{ bits/second} = 1.024 \text{ Mbits/second}$

3. Interface comparison

We compare our 1.024 Mbit/s requirement against the theoretical maximums of standard physical interfaces:

- SPI (50 Mbit/s): Yes, can sustain.
- I²C ($\leq 3.4 \text{ Mbit/s}$): Yes, can sustain.
- AXI4-Lite ($\sim 100 \text{ Mbit/s}$): Yes, can sustain.
- Decision: The lowest-complexity interface that comfortably handles 1.024 Mbit/s is I²C.

4. Burst analysis

Neuromorphic systems are highly asynchronous and prone to sudden bursts of activity.

- Burst Size: 25% of 1024 neurons = 256 spikes firing within a 1 ms window.
- Peak Instantaneous Rate: $256 \text{ spikes} / 0.001 \text{ seconds} = 256,000 \text{ spikes/second}$.
- Peak Bandwidth: $256,000 \times 20 \text{ bits} = 5,120,000 \text{ bits/second} = 5.12 \text{ Mbit/s}$.
- Burst-to-Mean Ratio: $5.12 / 1.024 = 5 \text{ times}$.
- Buffering Decision: The chosen I²C interface (max 3.4 Mbit/s) cannot absorb this 5.12 Mbit/s instantaneous peak. Buffering is absolutely required.
- Depth Estimate: In that 1 ms window, the chip generates 256 packets but I²C can only physically transmit 170 packets ($3.4 \text{ Mbit/s} \times 1 \text{ ms} / 20 \text{ bits}$). The FIFO buffer needs to be at least 86 packets deep (or $\sim 1.72 \text{ kbits}$) to prevent data loss.

5. Frame based comparison

What if we just used a standard dense clock-driven readout instead of event-driven AER?

- Frame Bandwidth: Sampling 1024 neurons every 1 ms (1000 Hz) at 1 bit/neuron = $1024 \times 1000 = 1,024,000$ bits/second = 1.024 Mbit/s.
- AER-to-Frame Ratio: $1.024 / 1.024 = 1.0$.
- Crossover Firing Rate ($f_{\text{crossover}}$): Since the bandwidths are exactly equal at 50 Hz, $f_{\text{crossover}} = 50$ Hz.
- Implication: Address-Event Representation (AER) is only the right choice for bandwidth efficiency when the neural network's average firing rate is below 50 Hz; if the neurons fire more rapidly, the AER packet overhead makes it less efficient than just sending a dense bit-vector.